

Quick Reaction Alert (QRA) is the state of readiness and modus operandi of air defence maintained at all hours of the day by NATO air forces. The United States usually refers to Quick Reaction Alert as Airspace Control Alert.

Some non-NATO countries maintain a QRA, either full-time or part-time.

Operation

QRA in the United Kingdom

There are two QRA stations in the United Kingdom, one at RAF Coningsby in the east of England, and the other at RAF Lossiemouth in Scotland.

Pilots and engineers on QRA duty are at immediate readiness twenty-four hours a day. They are fully dressed in the Crew Ready Room, which are next to the hangars, a hardened aircraft shelter known informally as Q-sheds, which houses the interceptor aircraft, since 2007 the Eurofighter Typhoon. Pilots are on QRA duty around once or twice a month, each a twenty-four-hour shift.

Engineers are on QRA duty three or four times a year, each for a twenty-four-hour a day shift, for seven days at a time. Two Typhoon aircraft are maintained at readiness, along with a Voyager tanker at RAF Brize Norton in Oxfordshire. Before 2014 this task was carried out by a TriStar.

Civilian aircraft in the UK are monitored by NATS Holdings at:

Swanwick, Hampshire with the London Area Control Centre and London Terminal Control Centre (at RAF West Drayton until 2007), broadly known as the London Air Traffic Control Centre, which covers the London Flight Information Region (EGTT). The RAF have a team at Swanwick.

Prestwick, which covers the Scotland Flight Information Region (EGPX) which is north of the

Solway Firth.

Military radar in the UK is controlled by the UK Air Surveillance and Control System (ASACS), looked after by ASACS Force Command. It has Remote Radar Heads (RRH) at:

RRH Benbecula in the Outer Hebrides

RRH Saxa Vord in Shetland (AN/TPS-77 radar moved from RRH Staxton Wold)

RRH Buchan (former Master Radar Station, or MRS, Buchan) off the A90 south of Peterhead

RRH Brizlee Wood in Hulne Park, Northumberland (west of RAF Boulmer)

RRH Staxton Wold off the B1249 in Willerby, North Yorkshire

RRH Neatishead at Neatishead, Norfolk

RRH Portreath in Cornwall

From 1997, the radars were the 1.1MW Plessey AR-320 (Type 93), and were replaced by the Lockheed Martin AN/FPS-117 system.

QRA in the United States

The United States refers to Quick Reaction Alert as Airspace Control Alert.

QRA response

Air traffic across Europe is controlled by Eurocontrol in Brussels. Military aircraft from Russia can be tracked across Norway, and reported to the Norwegian Joint Headquarters near Bodø, or the Combined Air Operations Centre 2 (CAOC UE) in Uedem, North Rhine-Westphalia close to the border with the Netherlands. Combined Air Operations Centre FINDERUP (CAOC FINDERUP), in Denmark, watches Russian aircraft and can alert the UK, and has RAF staff there. Russian Air Force Tupolev Tu-95 aircraft originate from the Olenya air base on the Kola Peninsula. Tupolev Tu-160 Blackjack aircraft come from the Engels-2 base near Saratov. The

Tu-95 aircraft are on 12-14 hour missions, and when tracked across Norway have been colloquially referred to with the codename of zombies.

A QRA response involves the fighter aircraft being scrambled to investigate an infringement of the NATO country's airspace or area of interest.

QRA response in the United Kingdom

This may also be a civilian aircraft that poses a threat, if not sufficiently responding to air traffic control (ATC). Incidents of this nature in the UK are monitored by the Control and Reporting Centre (CRC) at RAF Boulmer, which builds a 3D Recognised Air Picture. The National Air and Space Operations Centre (NASOC) at RAF Air Command, RAF High Wycombe decides whether to send a QRA response. The Joint Force Air Component Headquarters is also at High Wycombe.

QRA stations

Austria

Austria has only daylight QRA readiness. Austrian Air Force Air Surveillance Command is located at Salzburg. Fighter Squadron 1 & 2 with Eurofighter Typhoon are at Zeltweg Air Base.

Baltic States

Lithuania, Latvia and Estonia, though members of NATO, have no fighters capable of QRA intercepts. Other NATO nations provide periodic air defence at the NATO QRA standard as part of NATO Baltic Air Policing.

Belgium

Florennes Air Base and Kleine Brogel Air Base.

Bulgaria

Graf Ignatievo Air Base (3rd Fighter Air Base) of the BuAF has a single MiG-29 squadron, which carry two R-73 missiles. The Bulgarian CRC is at Sofia.

Croatia

The Croatian Air Force's Air Force and Air Defence Command is located in Zagreb. The 191. Fighter Squadron, flies Dassault Rafales from Pleso (Zagreb Airport).

Czech Republic

The 261st Control and Reporting Centre (CRC), is in Hlavenec. The Czech Air Force uses JAS-39C/D Gripen at Čáslav AFB.

Denmark

The Royal Danish Air Force Command is headquartered at Karup Air Base. The Fighter Wing with the Eskadrille 727 and the Eskadrille 730, at Skrydstrup Air Base use F-16AM/BM Falcon, soon to be replaced with F-35A.

Finland

The Finnish Air Operations Centre is at Jyväskylä-Tikkakoski Air Base. The Finnish Air Force Fighter Squadron 31 (Hävittäjälentolaivue 31, HävLLv 31) uses F-18C/D from Rovaniemi and Kuopio. In addition to the Air Force's main operating bases, QRA aircraft may be rotated between civilian airports and other temporary operating bases.

Germany

The municipality of Uedem houses NATO's Combined Air Operations Centre (CAOC) Uedem. Taktisches Luftwaffengeschwader 74 provides cover in the south, and Taktisches

Luftwaffengeschwader 71 "Richthofen" in the north. The German Air Force uses Eurofighter Typhoons from Wittmund (QRA North) and Neuburg Air Base (QRA South), with alternate QRA bases in Nörvenich and Rostock-Laage Airport.

Hungary

The Hungarian Air Command and Control Centre is in Veszprém. The QRA base is at the Kecskemét Air Base. The Fighter Squadron Puma operates JAS-39C/D Gripen fighters. The Hungarian Gripens are responsible for the air police service over Slovenia.

Iceland

Iceland, though a member of NATO, has no standing armed forces. Other NATO nations provide periodic air defence at the NATO QRA standard as part of NATO Icelandic Air Policing.

Italy

Italian Air Force squadrons equipped with Typhoon and F-35 jets provide QRA coverage on a rotational basis. These squadrons are based at Grosseto Air Base in Tuscany (IX Gruppo of 4° Stormo), Gioia del Colle Air Base in Apulia (10° Gruppo and 12° Gruppo of 36° Stormo), Trapani Air Base in Sicily (18° Gruppo of 37° Stormo), and Istrana Air Base in Veneto (132° Gruppo of 51° Stormo), all of them operating Typhoons. More recently the F-35As based at Amendola Air Base in Apulia (13° Gruppo of 32° Stormo) have been given QRA tasks as well, and F-35s at Ghedi Air Base in Lombardy will be on QRA too.

Netherlands

The Royal Netherlands Air Force (RNLAf) have F-35 aircraft at Volkel Air Base or Leeuwarden Air Base on high alert. They intercept once notified by the Air Operations Control Station Nieuw-Milligen, near Apeldoorn in Gelderland. The RNLAf alternates the responsibilities for

QRA above Benelux with the Belgian Air Component every four months since 2016/2017.

Norway

The Royal Norwegian Air Force Quick Reaction Alert force consists of two F-35s on high alert from Evenes Air Station.

Romania

The RoAF 71st Air Base (Baza 71 Aeriană) at Câmpia Turzii in central Romania and the RoAF 86th Air Base (Baza 86 Aeriană) at Borcea in south-east Romania are on QRA duty. Romania operates F-16 Fighting Falcons, which carry AIM-9X and AIM-120C missiles. The Romanian CRC is at Balotești in southern Romania.

Slovakia

The Air Force Command of the Slovak Air Force is in Zvolen. The 1st Tactical Squadron is waiting for delivery of F-16V at Sliač Air Base. Currently the Slovak airspace is guarded by German Air Force, Czech Air Force, Polish Air Force and Hungarian Air Force until the American F-16s are delivered.

Spain

The Spanish Air Force and Space Force establishes a number of multipurpose fighter units that take turns among themselves. On the one hand, there are four units dependent on the Air Combat Command (MACOM) that are responsible for protecting the airspace.

The QRAs are attended to by different predisposed units, which follow an action plan, in which several Air Force aircraft are prepared to respond to the alert within a maximum time of 15 minutes. A single base is not established as an operations center that is responsible for these activities, but there are several in Spanish territory that are "on duty" to respond to any

requirement.

Switzerland

The main base for the QRA of the Swiss Air Force is Payerne Air Base. The QRA also operates from Emmen Air Base and Meiringen Air Base for several weeks per year. Zurich Airport, Geneva Airport and Sion Airport serve as alternative locations. All QRA operations are guided by the Operations Center (EZ-LUV)/CRC at Dübendorf Air Base. The Swiss QRA are equipped with F/A-18C/D.

Turkey

Merzifon Air Base of the TuAF (Türk Hava Kuvvetleri), in northern Turkey, has one F-16 squadrons (built by TAI) with the 5th Air Wing (5 Ana Jet Üs). Bandırma Air Base has two F-16 squadrons of the 6th Air Wing. The Turkish CRC is at Ahlatlıbel near Ankara.

United Kingdom

Currently there are two QRA RAF stations, of 1 Group.

RAF Coningsby in Lincolnshire looks after the southern sector of the UK, known as QRA South, with 3 Sqn and 11 Sqn.

RAF Lossiemouth in Moray protects the northern sector of the UK, referred to as QRA North, hosted by 1 Sqn, 2 Sqn and 6 Sqn.

History

Quick Reaction Alert is the current iteration of scrambling, developed by RAF Fighter Command in the Battle of Britain. International boundaries were three miles off the coast.

Austria

On October 18 2002, a US DC-10 requested to enter Austrian airspace; it was intercepted by two Austrian SAAB fighter aircraft, but the aircraft turned out to be a McDonnell Douglas KC-10 Extender, with two Lockheed F-117 Nighthawk aircraft being refuelled.

Germany

RAF Wildenrath provided air defence cover for Royal Air Force Germany (RAFG), which flew Phantoms with 92 Sqn and 19 Sqn until 1991. The last QRA sortie at RAF Wildenrath was on Wednesday 2 October 1991.

Italy

The first country to put the Typhoon onto QRA duty was Italy in December 2005, by IX Gruppo of 4° Stormo. 12° Gruppo of 36° Stormo followed in July 2007 and 10° Gruppo in July 2010. Typhoons replaced the F-16A/ADF of 37° Stormo at Trapani from May 2012. From March 2018 the F-35 of 32° Stormo based in Amendola, has implemented the QRA assets.

Spain

111 Sqn put the first Typhoon on QRA duty in July 2008, followed by 142 Sqn of Ala 14, and later 141 Sqn.

United Kingdom

In the 1950s and 1960s, training as a fighter controller in the UK was at MRS Bawdsey (RAF Bawdsey). The main central control was known as ADOC, which monitored the UK Air Defence Region (UK ADR). It was similar to the USA's and Canada's NORAD at Peterson Air Force Base. The ROTOR system was developed in the 1950s.

Before computers arrived in the 1970s, the Russian aircraft were plotted on a map, mainly by

WRAF personnel. 11 Group at RAF Bentley Priory from 1968 and RAF High Wycombe from 1972, looked after the UK's air defence until the 1990s. High Wycombe today has the European Air Group. The Ilyushin Il-18 Coot would fly in the 1980s on Ferret missions over the North Sea. The Phantom would be in the air in under two minutes.

43 Sqn, the Fighting Cocks, operated the Phantom from Leuchars from September 1969, along with 23 Sqn, the Red Eagles and 11 Sqn, the Twin Eagles. 56 Sqn, the Firebirds, operated the Phantom in the 1970s from Suffolk. In 1979, an average of 5 Russian aircraft a week entered British airspace. The tankers at Marham were from 55 Sqn and 57 Sqn.

Every QRA alert required a Victor tanker from RAF Marham in Norfolk, with the codename Dragonfly. One fighter squadron would be on QRA for six month shifts. The Phantom had much better range than the Lightning, and had far-better look down radar, but the Lightning had better performance.

The RAF Phantom variant had Spey engines, which were not intentionally designed for the aircraft, and gave lower performance. It had an advanced jam-resistant inertial navigation system but the RAF Phantoms could not take off immediately as this inertial system had to align first. The Lightning left service in 1988 and the Phantom in 1992. When the Tornado F.3 arrived, the RAF QRA duty had an aircraft with complete night-vision capabilities and could connect to the Sentry aircraft.

In the 1960s, Southern Q was maintained by the Lightnings of 5 Sqn at RAF Binbrook and those of 29 Sqn and 111 Squadron at RAF Wattisham. Southern Q was rotated around the three RAF bases. The Lightning left QRA duty on Wednesday 27 April 1988, with last flight in July 1988.

RAF Leeming took over Southern Q from RAF Coningsby in 1988. 11 Sqn left RAF Leeming in October 2005. In June 2007, 3 Sqn at RAF Coningsby took over Southern Q from the Tornados of 25 Sqn at RAF Leeming in North Yorkshire. The Typhoons of 3 Sqn had their first scramble in August 2007 when they intercepted a Russian Bear. 3 Sqn took over all of Southern Q from

April 2008.

44 year old Pavel Grachev, the Minister of Defence (Russia) from 1992-96, was shown around RAF Leuchars on Thursday 23 July 1992, being shown the Tornado F3 QRA aircraft of 43 Squadron; he apparently enjoyed the 'hospitality' of the Scottish guests.

In August 2007 the Russians had begun to launch long-distance patrols after a 15-year hiatus. Typhoons arrived at RAF Leuchars with 6 Sqn from September 2010, performing their first QRA scramble in January 2011. Typhoons joined 1 Sqn from September 2012. 6 Sqn moved to Lossiemouth in June 2014, with 1 Sqn moving in August 2014. QRA North was moved from RAF Leuchars to RAF Lossiemouth in September 2014. The first QRA sortie from Lossiemouth was on 19 September 2014 with 6 Sqn.

2012 Olympics

To cover the security for the 2012 Summer Olympics, part of QRA South was briefly deployed from RAF Coningsby to operate from RAF Northolt.

Security incidents

1969 RAF Mildenhall C-130 theft, two French QRA aircraft took off

Wednesday 7 January 1976, a Lightning 'XP659' collided with a Piper Aztec 'G-BAJX' of Lease Air, travelling from Gothenburg, with pilot Arthur Ramsden, a former wartime bomber pilot, aged 52, and six people, at 600 ft; the Piper landed at Kirmington; the Aztec had been picked up as an unidentified aircraft

On 25 September 1979 a light aircraft 'G-BGHR' left at 1pm from Stansted airport; it descended from 31,000ft to 12,000ft, in an exercise at 2.36pm; it flew in circles, after the pilot became ill from hypoxia, and the plane began to make four-mile-wide circular paths for six hours, moving south, and crashed in a French vineyard at Le Tremblay at 8.20pm, 12 miles south-east of Nantes; the aircraft was followed by an RAF Nimrod over the English Channel,

and two French Air Force Mirage III and three Mirage F1 followed from 1810 to 2010; French pilots were given orders to shoot the plane down, if it was nearing a town; on the aircraft were Tom Lampitt, aged 47, from Loom Lane, Radlett in Hertfordshire, a former pilot of Aer Lingus, and Lieutenant-Colonel Fausto Aguiar de Barros Valla, from Sacavém in Portugal, of the Portuguese Army; Tom Lampitt, originally from New Penshaw, had flown in 43 Sqn at Leuchars in an aerobatics team with Peter Bairsto

1983 Scotland Learjet 25 crash, on Wednesday 18 May 1983, a pilotless Learjet 25 flew across the North Sea approaching Scotland, crashing 260 miles off Stornoway at around 8.30pm; a Nimrod from RAF Kinloss searched for wreckage; the West German aircraft had taken off from Vienna International Airport at 2.53pm, flying at around 40,000 ft (FL390) across Holland; radio contact had been lost near Fulda in West Germany at 3.49pm; the Learjet had been followed by Dutch QRA pilots, who had likewise seen no-one at the controls; the aircraft 'D-CDPD' was registered by Air Traffic GmbH of Munich; the pilot was believed to be Peter Blangman; the Learjet was intercepted by an RAF Phantom over Loch Rannoch, with pilot 23 year old Flying Officer Mark Hanna, the son of Ray Hanna, and navigator 28 year old Flt Lt John Marr, of Portsmouth, from 111 Sqn at Leuchars; the RAF aircrew could see no-one in the cockpit, from a distance of 30 feet; the Learjet flew at around 450mph: the Phantom followed the aircraft to around 150 miles west of Kinloss, as there was no tanker aircraft; from there the Learjet was followed by another QRA RAF Phantom from RAF Coningsby; it crossed the Outer Hebrides at 5.30pm

1989 Belgium MiG-23 crash, on 4 July 1989, where multiple QRA aircraft were scrambled
Helios Airways Flight 522, on 14 August 2005, intercepted by two Greek F-16 aircraft from Nea Anchialos National Airport

2014 Olsberg mid-air collision, on 23 June 2014, a QRA exercise resulted in a mid-air collision between a Learjet 35 and a German Air Force aircraft from Nörvenich Air Base

2018 Horizon Air Bombardier Q400 incident, on 10 August 2018 in the western US

2022 Baltic Sea Cessna Citation crash

Non-NATO security incidents

Ryanair Flight 4978 in May 2021, ordered to divert from the scheduled route

1987 Okinawan Tu-16 airspace violation

Defection of Viktor Belenko

Mathias Rust, landed on the Bolshoy Moskvoretsky Bridge on 28 May 1987 aged 18, in his Cessna 172, flying from Uetersen Airfield in Germany, via Helsinki-Malmi Airport; a MiG-23 intercepted him, but the Soviet fighter pilot was told to leave the aircraft alone. The Soviet Air Defence Forces, known as the PVO, had given the light aircraft an incorrect IFF identification, and the aircraft was also misidentified as a low-flying helicopter, which was likely of no defence importance. It was also the annual Border Guards Day, which provided much distraction.

1960 RB-47 shootdown incident

Violations of Japanese airspace

See also

Index of aviation articles

List of established military terms

ACCS

Baltic Air Policing

Cold War II

Minimum interval takeoff

NATO Integrated Air Defense System (NADGE, or NATO Air Defence Ground Environment, in the 1950s)

Category:Military radars of the United Kingdom

Category:Telecommunications equipment of the Cold War

Category:Soviet Long Range Aviation bases

References

External links

September 2012 MoD document

"Quick Reaction Alert" (PDF). RAF 2004 (Report). Royal Air Force. 2004. pp. 38–43. Retrieved 3 August 2015.